Team 7

Traffic Congestion Level Prediction

Using Random Forest & Gradient-Boosted Trees

Team Members:
N. Greeshma | 2520030080
E. Nakshatra | 2520030090

Machine Learning | Section 08
Guide : Shaik Asif
Assistant Professor CSE

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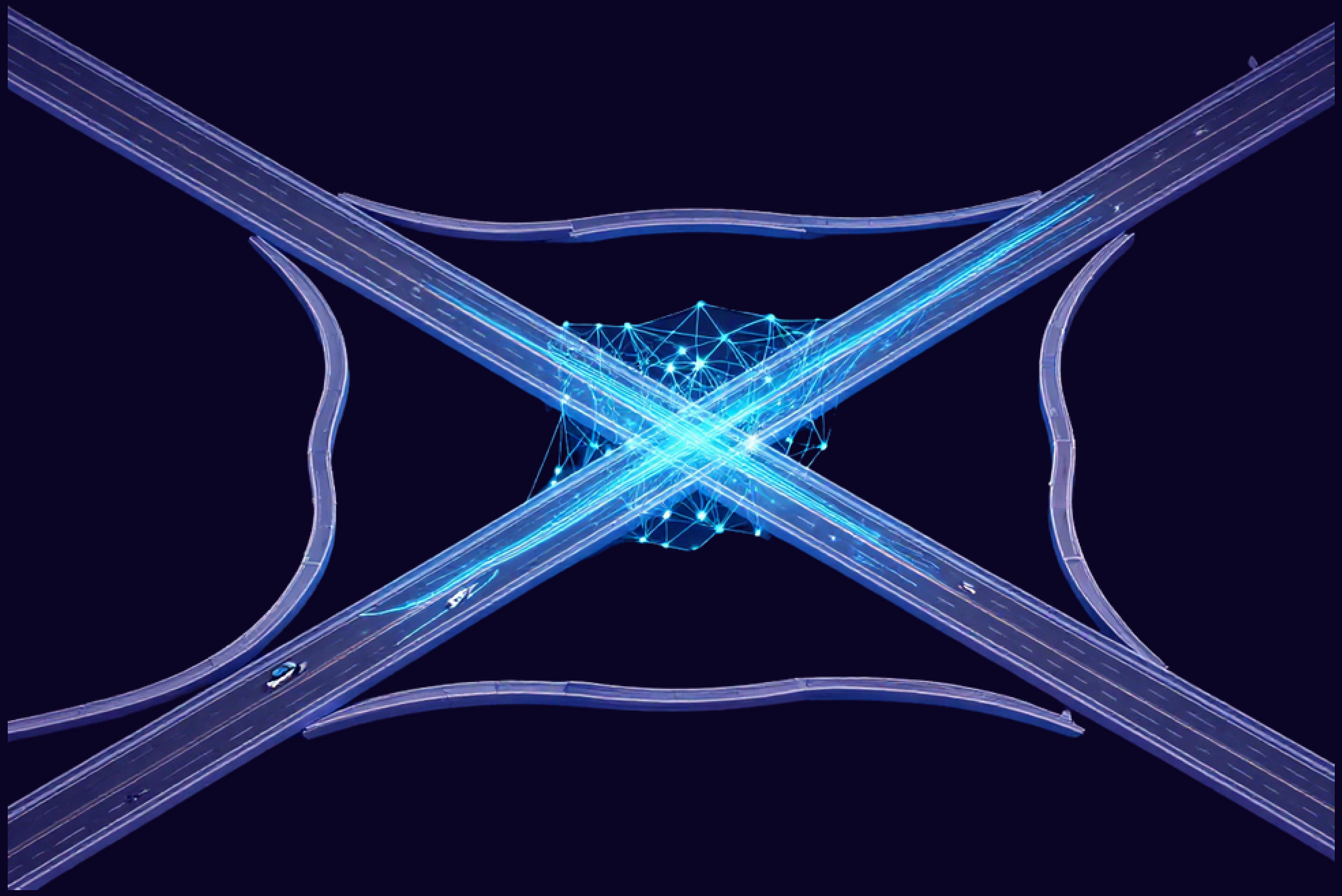


Problem Statement

- Increasing vehicles and limited road infrastructure cause severe traffic congestion.
- Congestion results in:

1. Increased travel time
2. Fuel consumption
3. Air pollution
4. Economic losses

- Traditional systems mainly monitor traffic, rather than predicting future congestion.

Our Goal:
Predict traffic congestion levels using Machine Learning.

Solution

Our proposed system uses Machine Learning to predict traffic congestion levels by analyzing multiple traffic and environmental factors.

Input Factors

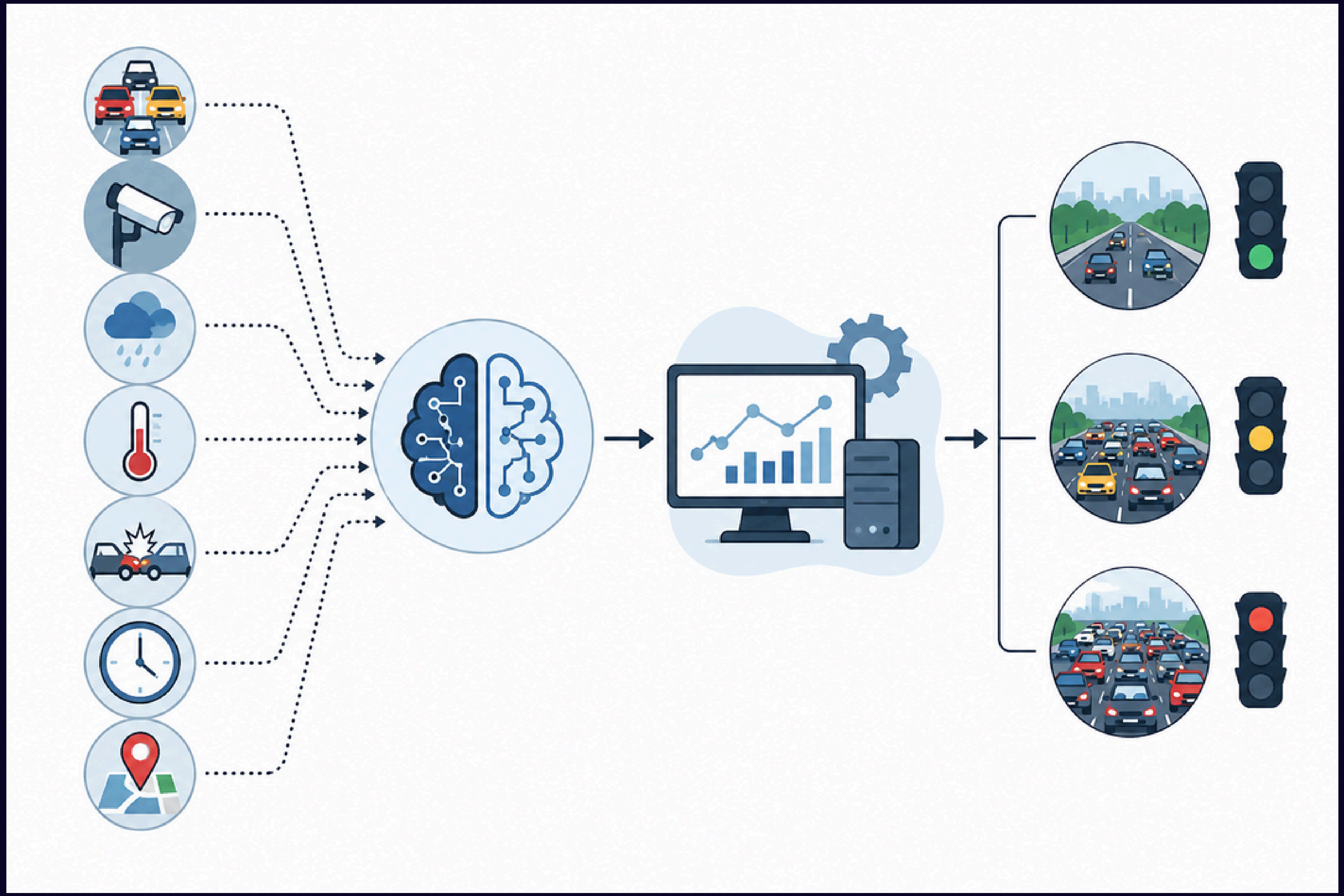
- Traffic Volume
- Average Vehicle Speed
- Vehicle Counts
- Weather Conditions
- Temperature & Humidity
- Accidents
- Time & Location
- Traffic Signal Status

Machine Learning Models

- Random Forest
- Gradient-Boosted Trees

Output

🚦 Traffic Congestion Level
Low | Medium | High



Compare both models and identify the better-performing approach for traffic congestion prediction.

Objectives

- Predict traffic congestion levels accurately.
- Analyze important traffic and environmental factors.
- Implement Random Forest and Gradient-Boosted Trees.
- Compare the performance of both algorithms.
- Develop a system that can support better traffic planning and management.

Dataset Overview

Smart Traffic Management Dataset
Source: Kaggle

Features:

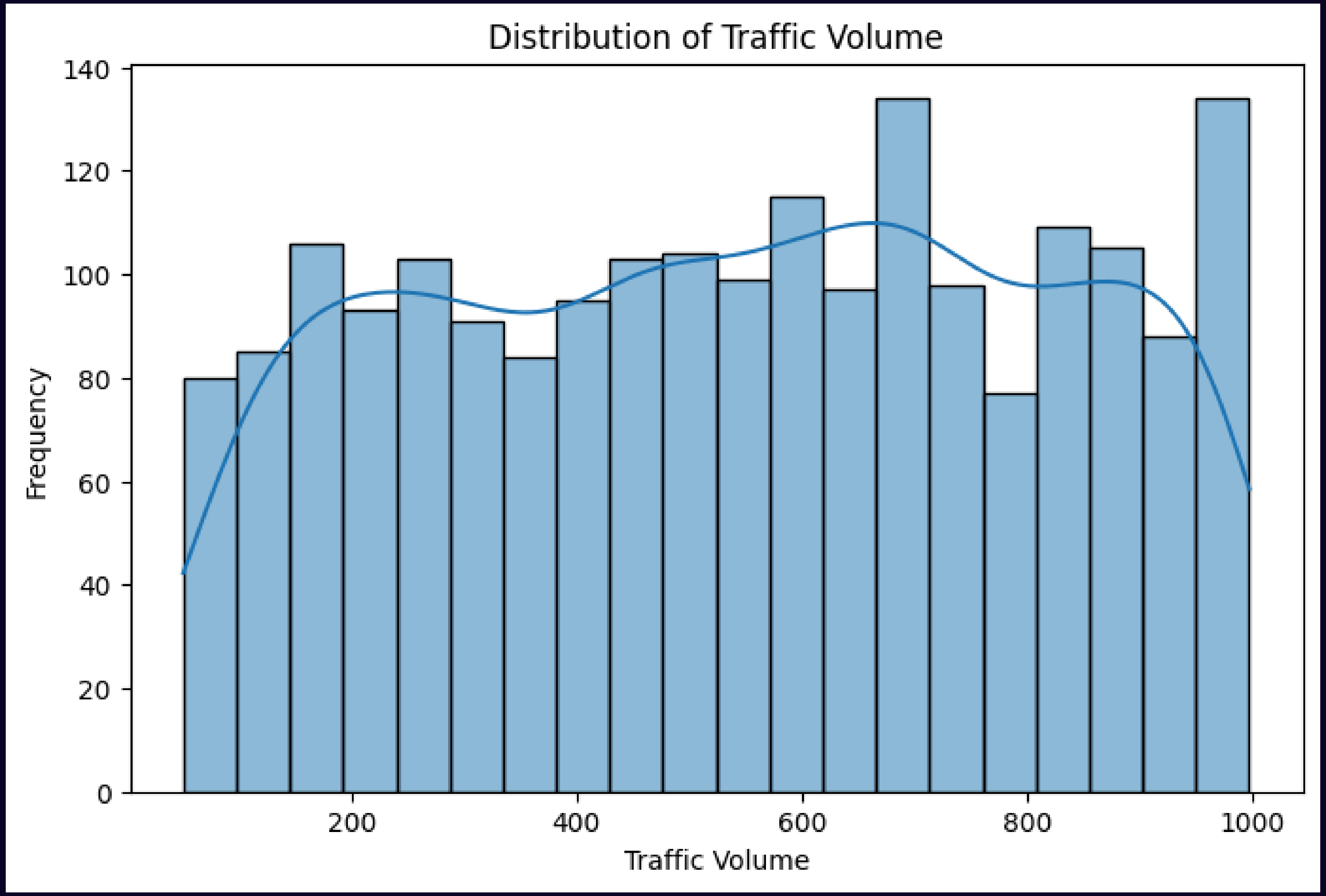
- Timestamp
- Location ID
- Traffic Volume
- Average Vehicle Speed
- Cars
- Trucks
- Bikes
- Weather Condition
- Temperature
- Humidity
- Accident Reported
- Signal Status

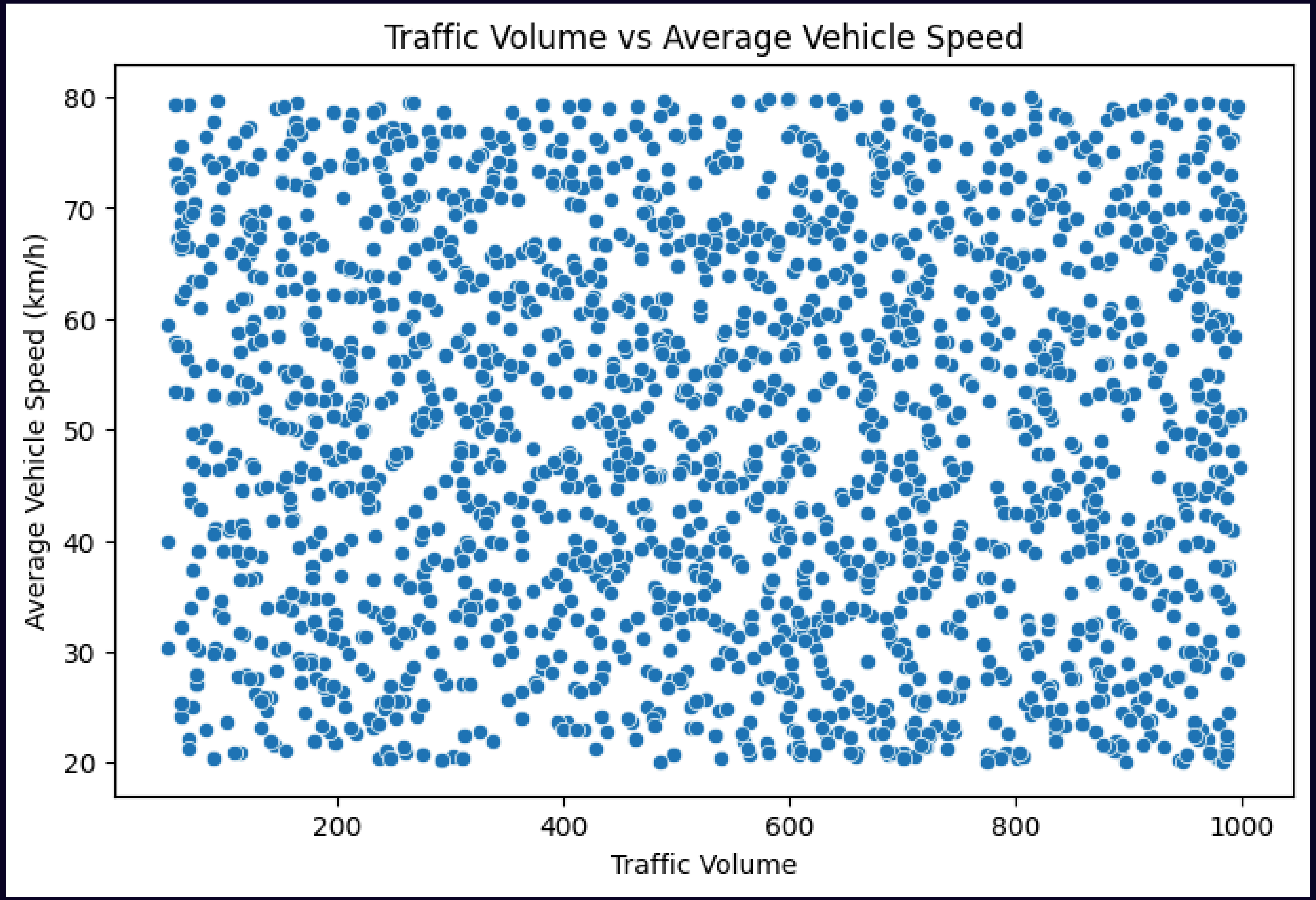
Dataset Size:

- 2,000 rows
- 12 columns

Exploratory Data Analysis (EDA)

EDA was performed to understand the dataset, identify patterns, check data quality, and determine relationships between traffic-related variables.

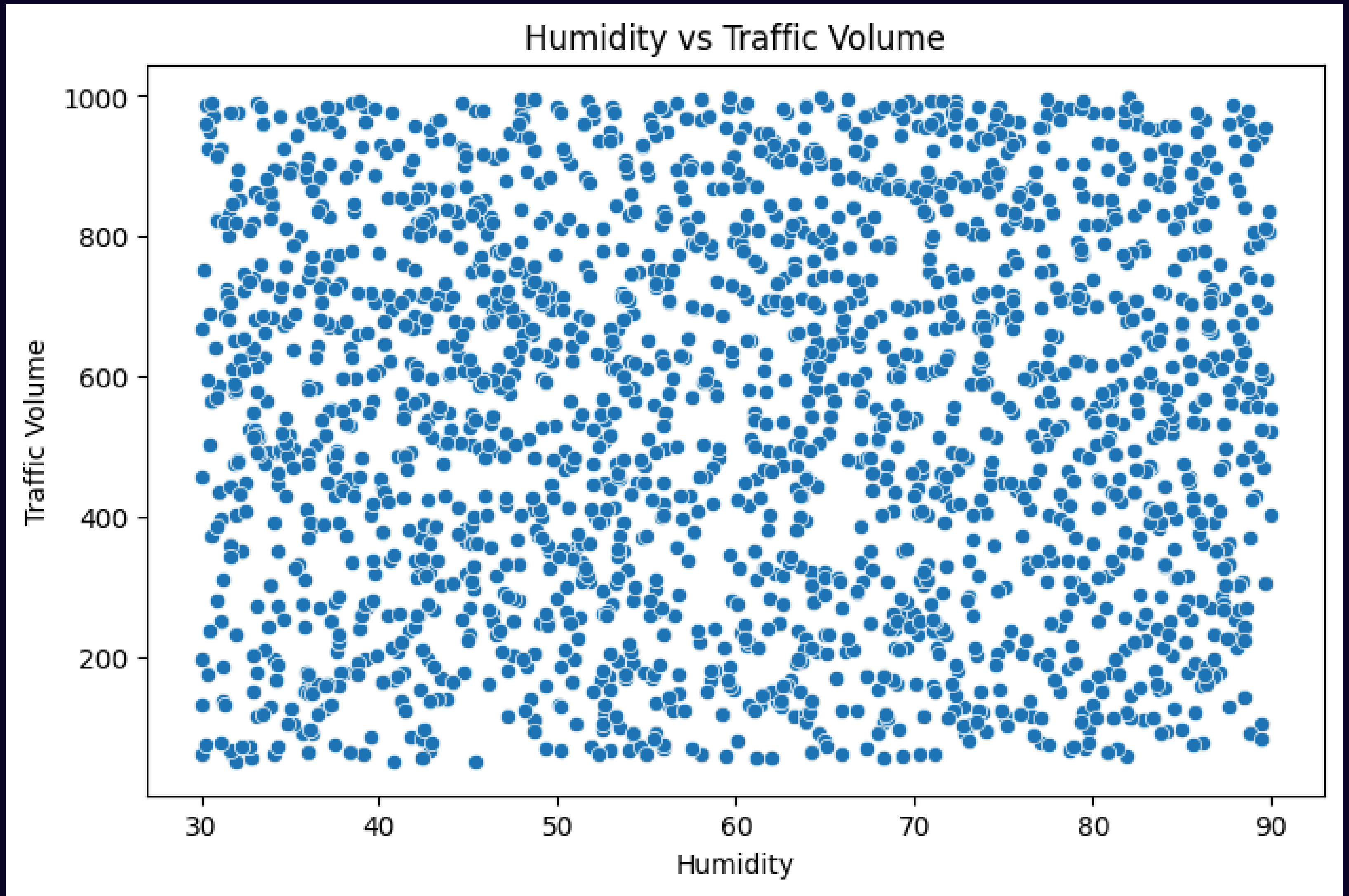


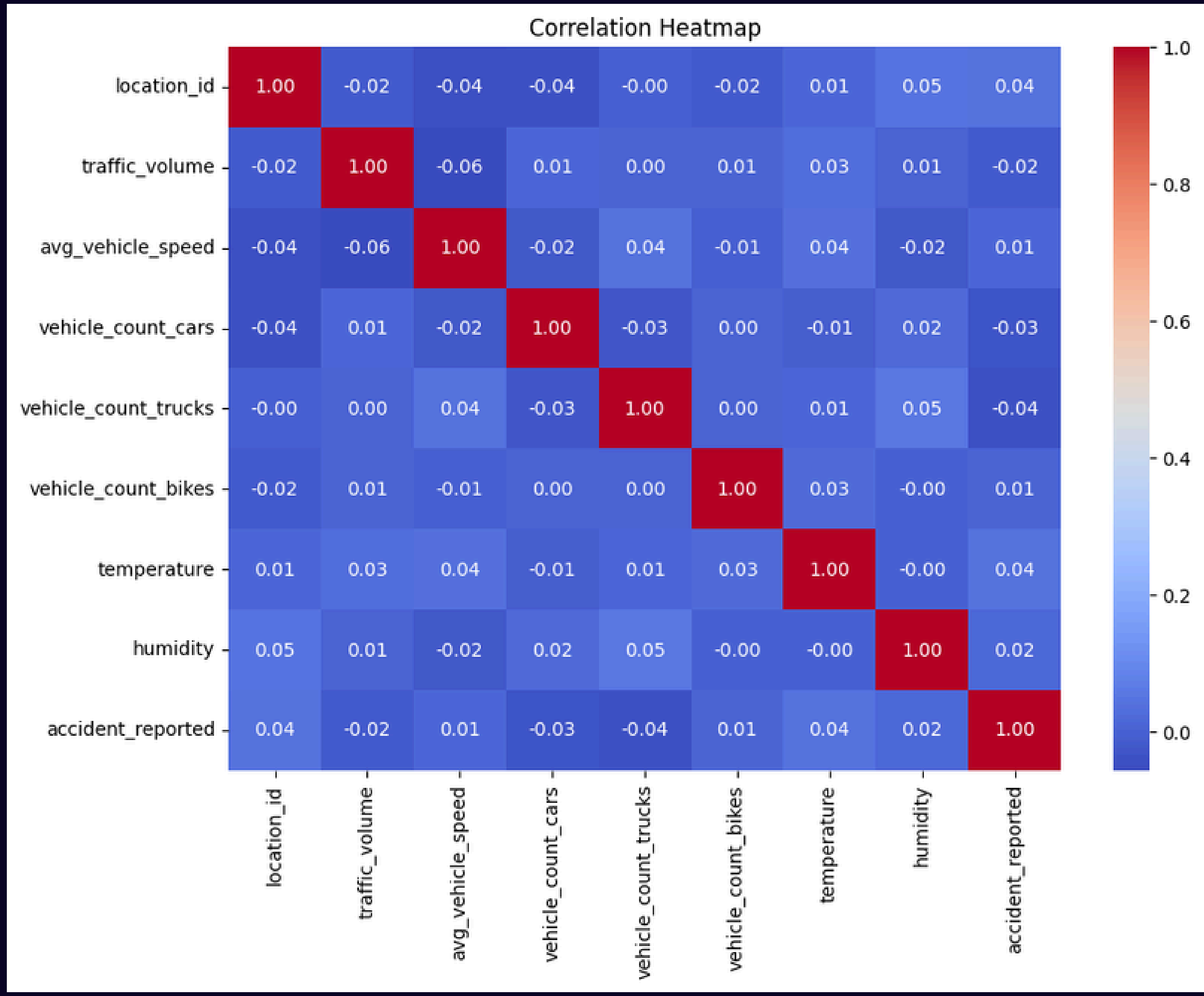


EDA Findings:

Some important observations:

- Traffic volume is distributed across a wide range.
- Average vehicle speed ranges approximately from 20–80 km/h.
- Traffic volume and speed do not show a strong linear relationship.
- Traffic volume varies across different weather conditions.
- Multiple factors influence traffic conditions, so using multiple features is important.





Research Papers / Literature Survey

Research Paper	Algorithm	Key Finding
Traffic Congestion Prediction Based on ETA	Random Forest	Random Forest showed effective congestion classification.
Ensemble Learning for Short-Term Traffic Prediction	Gradient Boosting	Gradient Boosting performed well for traffic prediction.
Road Traffic Flow Estimation Using ML	Random Forest + XGBoost	Ensemble models effectively captured complex traffic patterns.

Previous studies show that ensemble learning methods such as Random Forest and Gradient Boosting are effective for traffic prediction, supporting our choice of algorithms.

Conclusion

- Traffic congestion is a major urban problem that requires accurate prediction and better management.
- Our project uses Machine Learning to classify congestion levels as Low, Medium, or High.
- Random Forest and Gradient-Boosted Trees are used to analyze complex traffic patterns.
- EDA helps us understand the influence of traffic, speed, weather, and environmental factors.
- The performance of both models will be evaluated and compared to identify the better approach.
- The proposed system can support smarter traffic planning and congestion management.

Our goal is to turn traffic data into actionable predictions for smarter and more efficient transportation.

Traffic Congestion
Level Prediction

Thank You

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Random Forest & Gradient-Boosted Trees